

Assignment Questions-Module 1

1. In which OSL layer the Wi-Fi standard/protocol fits

- **Physical Layer (Layer 1)** → Radio transmission (frequency, modulation)
- **Data Link Layer (Layer 2)** → MAC addressing, frame control, error handling

2. Can you share the Wi-Fi devices that you are using day to day life, share that device's wireless capability/properties after connecting to network. Match your device to corresponding Wi-Fi Generations based on properties

Device	Standard	Band	Max Speed	Wi-Fi Generation
Smartphone	802.11ac	5 GHz	~866 Mbps	Wi-Fi 5
Laptop	802.11ax	2.4/5 GHz	~1200 Mbps	Wi-Fi 6

3. What is BSS and ESS?

A **single Wi-Fi network** with one Access Point (AP) which is identified by **BSSID (MAC address)** Example: Your home router network

Multiple APs connected (same SSID) which allows **roaming without disconnection**
Example: College campus Wi-Fi

4. what are the basic functionalities of Wi-Fi Access point

functions include:

- Providing wireless connectivity to multiple clients
- Acting as a bridge between wireless and wired networks
- Managing authentication and security protocols such as WPA2/WPA3
- Assigning IP addresses through DHCP (in home routers)
- Controlling data traffic and ensuring efficient communication
- Supporting roaming in extended networks

The AP ensures reliable and secure communication between devices and the internet.

5. Difference between Bridge mode and Repeater mode

In **Bridge Mode**, a router connects two separate networks, typically using a wired connection. It disables routing functions like NAT and allows devices to be on the same network. This mode provides high performance and minimal signal loss.

In **Repeater Mode**, a device wirelessly connects to an existing network and rebroadcasts the signal to extend coverage. However, since it uses the same channel for receiving and transmitting, it reduces the overall bandwidth and speed.

6. what are the difference between 802.11a and 802.11b

802.11a operates in the **5 GHz band** and supports speeds up to **54 Mbps**. It experiences less interference because fewer devices use this band, but it has a shorter range due to higher frequency.

802.11b operates in the **2.4 GHz band** and supports speeds up to **11 Mbps**. It provides better coverage and wall penetration but suffers from higher interference due to other devices like Bluetooth and microwaves.

7. Configure your modem/hotspot to operate only in 2.4Ghz and connect your laptop/Wi-Fi device and capture the capability/properties in your Wi-Fi device. Repeat the same in 5Ghz and tabulate all the differences you observed during this

When configuring a Wi-Fi router to operate in **2.4 GHz**, the network provides wider coverage and better penetration through walls. However, it suffers from interference and lower data speeds due to limited channels and congestion.

When configured to **5 GHz**, the network offers higher data rates, more channels, and less interference. However, the signal range is shorter, and it cannot penetrate obstacles effectively.

8. What is the difference between IEEE and WFA

The **IEEE (Institute of Electrical and Electronics Engineers)** is an international organization that develops technical standards such as the 802.11 Wi-Fi protocols. It defines how wireless communication should function.

The **Wi-Fi Alliance (WFA)** is an industry consortium that certifies devices based on IEEE standards. It ensures interoperability between devices from different manufacturers and provides the “Wi-Fi CERTIFIED” label.

9. List down the type of Wi-Fi internet connectivity backhaul, share your home/college's wireless internet connectivity backhaul name and its properties

- **Fiber Backhaul:** Provides high speed, low latency, and high reliability
- **Ethernet Backhaul:** Wired connection used in mesh systems
- **Wireless Backhaul:** Used in mesh networks without cables
- **DSL/Cable:** Moderate speed using telephone or cable lines
- **Cellular (4G/5G):** Used in mobile hotspots

10. List down the Wi-Fi topologies and use cases of each one.

Infrastructure Mode is the most common, where devices connect through an access point. It is used in homes and offices.

Ad-hoc Mode allows direct communication between devices without an AP. It is used for quick file sharing.

Mesh Topology consists of multiple interconnected APs that provide seamless coverage. It is used in large areas like campuses and smart homes.

Point-to-Point Topology connects two locations directly, often used between buildings.

Point-to-Multipoint Topology connects one central node to multiple endpoints, commonly used by internet service providers.